

Section 6.1: Negative Aspects of ICT

ICT can affect our lives in both positive and negative ways. In this section we take a look at some of the problems that the use of ICT can bring.

Some of the areas we will cover includes copyright problems, viruses and hacking.

Key Concepts of this section:

- # Understand the issues with **software copyright theft**
- # Know the problems that **computers viruses** can cause and how to prevent them
- # Be able to discuss what is meant by '**hacking**' and how to prevent it

Software Copyright Theft

Definition: *"Downloading, copying or borrowing software (for free) without the permission of the owner "*



Key Words:

Copyright, Piracy, Serial Numbers, Hologram, Licensing

Overview of software copyright theft:

- # **Software** is often **developed at massive costs** (both financially and in terms of time) by companies and individuals.
- # Because software is digital, it is very **easy to either download illegally** from the internet or to **make copies of it** without the permission of the owner. (i.e. copying a CD or downloading from piratebay). This is often called '**Software Piracy**'.
- # Laws exist to help prevent the developers of software from having their work downloaded or copied by others for free.

Click for Example of
Potential Costs



What do these laws cover?

- 1 Software should **not be copied, sold or given away** to other people
- 2 Software should **not be used on a network of more users than you have paid** for (e.g. 20 users instead of 10)
- 3 **Renting out software** without permission is **illegal**
- 4 **Stealing code from software** and then using it in your own software product **is illegal**
- 5 Using the **name of copyright software on your product is illegal** (Passing off your product as Microsoft Word for example)

Software Piracy Prevention

Software can be protected against piracy in a number of ways including:

- #. 1. **Serial Numbers**
- #. 2. **Holograms**
- #. 3. **Licence agreements**

1 Serial Numbers

When software is being installed, users are often asked to enter a **unique number which proves the software is original**.

Serial numbers are supplied with the original copy of the software.



2 Holograms

Holograms are often used on the packaging of software and they indicate that the **copy is genuine**.

Software without a hologram on the packaging is usually an illegal copy.



3 Licence Agreements

Licence agreements are part of the software installation process and they inform the user exactly how they are **legally allowed to use the product**.

Users are required to **'agree to terms of use'** before they can complete the installation.



Computer Viruses

Definition: *"A self-copying program that has been created with the purpose of damaging a computer system"*

Overview of computer viruses:

- # Computer viruses can cause huge problems if they are allowed to infect a computer.
- # Viruses are programs that automatically make **replicates (copies) of themselves** and then **attach to other files** within the computer.
- # Any files which have viruses attached to them are said to have been '**infected**'.

What problems can viruses cause?

- # Viruses can be designed to carry out different malicious (harmful) functions. Some include:
 - **Annoying pop-up messages on the screen**
 - **Causing the computer to slow down or even crash (stop responding)**
 - **Deleting important files or programs**
 - **Trying to access your personal files like passwords, bank details or credit card numbers etc**



Key Words:

Virus, Replicate, Infect, Crash, Pop-Up, Prevention



The first computer virus sent over the Internet was created in 1988 by Robert Morris.

[Click for more info](#)

How do viruses infect computers?

There are many ways that viruses can infect a computer. The most common include:

- **Downloading infected files from the internet**
- **Email attachments that contain infected files**
- **Plugging a flash memory stick that containing infected files into your computer**

Preventing Software Virus Infections

New computer viruses are being created all the time and this means that you should take steps to reduce the chance that you computer and data become infected. The best methods are outlined in the table below:

- 1 Use **up-to-date anti-virus software**.
- 2 Use **genuine copies** of software.
- 3 Only open **email attachments from people you trust**.
- 4 **Scan flash memory with anti-virus software** before you attach it to your computer.
- 5 Only download software from **legitimate websites**.



Click for Picture
Example

NOTE!

Keeping your **anti-virus software up-to-date** is vital. This is because new versions of viruses are being made all the time.

Hacking

Definition: *"The process of accessing a computer system without permission"*



Key Words:

Virus, Infect, Crash, Pop-Up, Prevention

Overview of hacking:

Hacking is where people '**break**' into computer systems without permission. These people are known as '**hackers**'.

Hackers often use the **Internet** to break into the system.

Why do people try to hack into computer systems?

Hackers try to break into computers for the following reasons:

- **To cause damage to files and data by deleting or changing them**
- **To commit fraud by stealing data (bank numbers etc)**
- **To access sensitive information**
- **Simply to see if they are clever enough to beat the system's security.**



NOTE!

Some companies actually **employ hackers** as a way of **testing how secure their security is**.

Preventing Hacking and Hackers

There are various methods you can use to help stop hackers from accessing your computer files and data. These include:

- #. **1. Firewall's**
- #. **2. User-ID's and Robust Passwords**
- #. **3. Data Encryption**

1 Firewall's

Firewall's provide a '**shield**' between your **computer** and the **Internet**.

Firewall's can **block unwanted data** (including hackers) from reaching your computer. They can also stop your computer from connecting to unwanted websites.



2 User-ID's and Robust Passwords

U User-ID's

A user-ID is usually a **word or a number** that identifies particular users as they log onto the computer system.

User-ID's give people access to certain areas or files within the computer.

U Robust Passwords (hard to guess)

A good password should be hard to guess. It should be made up of a **combination of letter and numbers** and changed regularly.

NOTE!

In order to gain access to the system, the **hacker** must



guess/break both the User-ID AND the Password.

The **more robust the password** is, the **harder it will be for the hacker.**

3 Data Encryption

Encryption makes the **files on you system unreadable** if the user does not know the password to un-encrypt them.

If someone try's to read an encrypted file they will just see **random junk instead of the real data.**

NOTE!

Encryption **does not prevent hackers** from **accessing your files** it just makes them **unreadable and useless.**



Activity!

[Negative Aspects of ICT - Questions](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all questions in full.



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